

DB-2.0

Definition Breaker

Every Child Directs the World

A child-operable accessibility standard, multisensory learning path, digital game design, physical-toy scaffold and printable play system.

WHERE?

MATCH?

WHEN?

CHECKED?

CHANGE

REMEMBER

I DECIDE

Choose

look, listen, touch, point, speak, sign, scan, press or ask a partner

Control

stop, wait, repair, switch route, take a break and return

Learn

the real substrate ideas in child words, builder words and formal words

Executive decision: the child is the authority

Accessibility is not a mode. It is a lawful route through the world.

DB-2.0 treats accessibility as part of the same rule system that governs the game. Access does not sit outside play, lower the learning goal or hand the child's decision to an adult.

1

The child selects the route

A route may combine visual, auditory, tactile, motor, communication, language, regulation and energy supports. The child may change it during play.

2

The same idea remains

Access may change channel, pace, layout, input or explanation. It does not secretly replace support, compatibility, guards, verified change or lineage with a lesser task.

3

Assistance is valid play

AAC, switches, gaze, co-play and partner-assisted selection count when the child retains choice, confirmation, repair and the right to stop.

4

Exclusion blocks release

A single-sense clue, drag-only action, inaccessible help, forced touch, unconfirmed partner choice or lower accessible reward is a product defect.

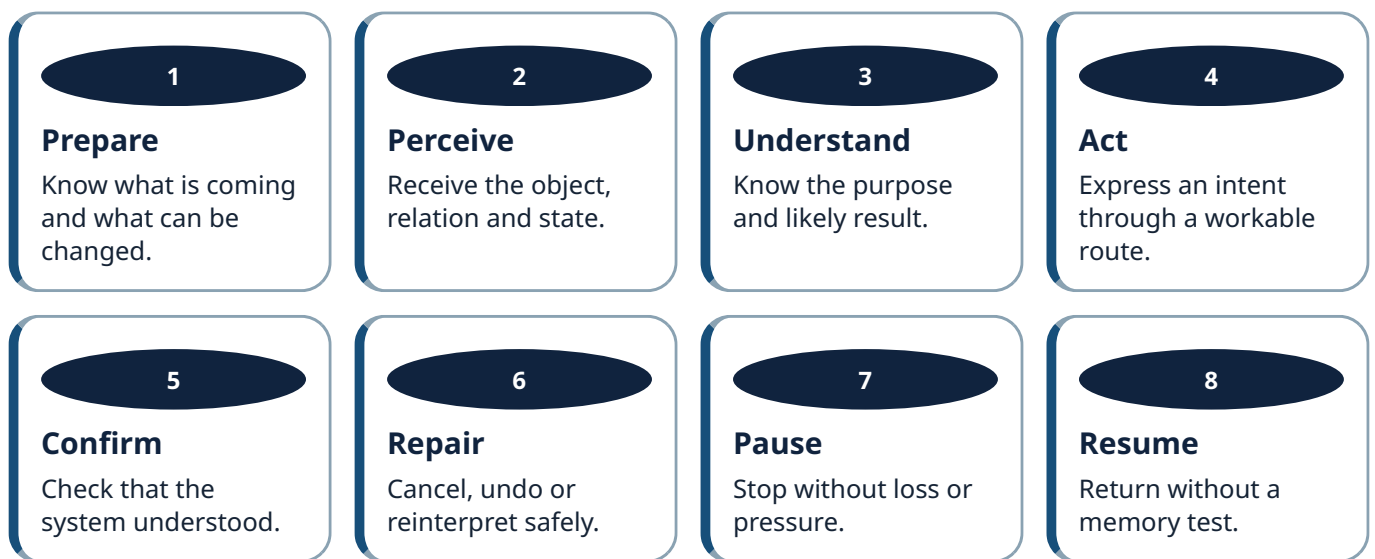
DB-2.0 decision

GO as the new child-access baseline. Every essential action must work through a complete access route, preserve agency, and return to the same verified world-change pipeline.

The eight-stage child access envelope

Prepare - Perceive - Understand - Act - Confirm - Repair - Pause - Resume

An action is accessible only when the whole journey works. Being able to activate one control is not enough.



Parity rule

Different access routes must preserve meaning, outcome, dignity, reward, child control and reasonable effort.

How to use this edition

No reader has to read the whole book. Choose the path that helps today.

A

A child or peer

Begin at Part A. Read one page, try one action, stop whenever needed. The pages are designed to be relayed by speech, sign, AAC, touch or object cue.

B

Family or facilitator

Use Parts B, C and G. Support communication, offer real choices, wait, confirm and repair. Do not turn access into a compliance test.

C

Game or learning designer

Use Parts D and E to map child-facing language onto the actual substrate and implementation contracts.

D

Toy or tabletop designer

Use Part F and the printables. The physical concepts are prototypes, not certified manufacturing drawings.

Reading is not a test

A reader may magnify, listen, print, skip, receive a tactile explanation, use a partner, or return later. Eye contact, speech, speed and independent hand use are never prerequisites.

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My rights in this world

I am the player. My way of communicating counts.

STOP

I can stop

STOP works before, during and after an action. I do not have to explain why.

CHOOSE

I can choose

I can look, listen, touch, point, speak, sign, scan, press or use a partner.

WAIT

I can wait

The game does not decide that I failed because I need more time.

REPAIR

I can repair

I can say NOT THAT, undo, change route or try again.

BREAK

I can rest

A break does not erase my work or turn into a memory test.

KNOW

I can know

The game tells me what will happen, what changed and what it stores.

Adult promise

Adults and systems must notice, honor and confirm these rights. Silence, stillness, delay or unusual movement are not automatic consent.

You lead the game

The world waits for your choice. A helper may explain or move a piece, but the choice stays yours.

1

Step 1

Choose a rule card.

2

Step 2

Choose how the card is shown.

3

Step 3

Confirm before the world changes.

Try it

Try: point to, look at, touch, name, sign or scan to one card. Then use YES or NOT THAT.

You can stop

STOP, WAIT, HELP, NOT THAT and BREAK remain available on every page.

STOP works now

STOP must work from every screen, room, toy activity and partner interaction. It does not need a reason.

1

Step 1

STOP pauses action.

2

Step 2

The current choice stays safe.

3

Step 3

The next step is explained when you return.

Try it

Try: show your STOP signal. The adult or game stops immediately and says what is paused.

You can stop

STOP, WAIT, HELP, NOT THAT and BREAK remain available on every page.

HELP keeps my progress

HELP means “make the route work,” not “take over my decision.”

1

Step 1

Help can explain.

2

Step 2

Help can reduce clutter.

3

Step 3

Help can offer another input route.

Try it

Try: ask for HELP, then choose which kind of help you want.

You can stop

STOP, WAIT, HELP, NOT THAT and BREAK remain available on every page.

Choose how to play

You may combine routes. You do not have to fit one fixed access mode.

1

Step 1

Look or magnify.

2

Step 2

Listen or read captions.

3

Step 3

Touch, scan, gaze, press, sign or use a partner.

Try it

Try: select two routes that feel useful today.

You can stop

STOP, WAIT, HELP, NOT THAT and BREAK remain available on every page.

Look, listen or touch

The same important rule can be shown with words, shape, border pattern, spoken name and tactile mark.

1

Step 1

No color-only clue.

2

Step 2

No sound-only clue.

3

Step 3

No picture-only instruction.

Try it

Try: cover one channel. Can you still find the rule?

You can stop

STOP, WAIT, HELP, NOT THAT and BREAK remain available on every page.

I can answer in many ways

A real answer can be speech, sign, AAC, point, gaze, switch, body movement or partner-confirmed signal.

1

Step 1

The system waits.

2

Step 2

The system confirms.

3

Step 3

The system allows repair.

Try it

Try: answer YES in one way and NOT SURE in another.

You can stop

STOP, WAIT, HELP, NOT THAT and BREAK remain available on every page.

YES, NO and NOT SURE

NOT SURE is a complete answer. It is not a hidden NO and not a mistake.

1

Step 1

YES chooses.

2

Step 2

NO rejects.

3

Step 3

NOT SURE asks for more information or time.

Try it

Try: ask what a rule will do before choosing YES or NO.

You can stop

STOP, WAIT, HELP, NOT THAT and BREAK remain available on every page.

I changed my mind

A choice can be changed before commit. After commit, undo or replay can support repair when the puzzle allows it.

1

Step 1

Changing your mind is information.

2

Step 2

It is not cheating.

3

Step 3

It does not reduce reward.

Try it

Try: choose one tile, say CHANGED MIND, then choose another.

You can stop

STOP, WAIT, HELP, NOT THAT and BREAK remain available on every page.

Wait for me

The game should separate thinking time from challenge time.

1

Step 1

No disappearing choices.

2

Step 2

No repeated prompting without permission.

3

Step 3

No score penalty for access time.

Try it

Try: use WAIT. The current focus and choices stay where they are.

You can stop

STOP, WAIT, HELP, NOT THAT and BREAK remain available on every page.

Quiet, dim or still

You may reduce sound, brightness, motion, particles, vibration or visual clutter without losing information.

1

Step 1

Quiet keeps captions.

2

Step 2

Still keeps state changes clear.

3

Step 3

Dim keeps contrast readable.

Try it

Try: choose one sensory change before the room starts.

You can stop

STOP, WAIT, HELP, NOT THAT and BREAK remain available on every page.

First, then, next, break, finished

A short schedule makes the activity predictable without locking you in.

1

FIRST

FIRST: notice the room.

2

THEN

THEN: choose one rule.

3

NEXT

NEXT: preview. BREAK and FINISHED stay available.

Try it

Try: move a marker along the schedule. You may move it back.

You can stop

STOP, WAIT, HELP, NOT THAT and BREAK remain available on every page.

Know before the room starts

The game tells you about sound, light, motion, touch, waiting, reading and teamwork before play.

1

Step 1

Expected intensity is named.

2

Step 2

Surprises are optional.

3

Step 3

A safer route is offered.

Try it

Try: choose PLAY, CHANGE SETTINGS or SKIP.

You can stop

STOP, WAIT, HELP, NOT THAT and BREAK remain available on every page.

One thing at a time

A focus view can hide nonessential detail while keeping the full idea.

1

Step 1

One current object.

2

Step 2

One clear question.

3

Step 3

One next action.

Try it

Try: ask for LESS ON SCREEN. The room does not reset.

You can stop

STOP, WAIT, HELP, NOT THAT and BREAK remain available on every page.

One-button route

One button can move through choices and select without rushing.

1

Step 1

Items highlight in a stable order.

2

Step 2

The child controls select.

3

Step 3

Pause and back remain reachable.

Try it

Try: let the highlight move. Press only when your choice is named.

You can stop

STOP, WAIT, HELP, NOT THAT and BREAK remain available on every page.

Two-button route

One button can move and one can choose. Both actions can be remapped.

1

Step 1

NEXT changes focus.

2

Step 2

SELECT confirms focus.

3

Step 3

A long hold is not required.

Try it

Try: move through three rule cards and select one.

You can stop

STOP, WAIT, HELP, NOT THAT and BREAK remain available on every page.

Step-scan route

The child advances the scan one step at a time and can rest without the scan moving away.

1

Step 1

No automatic race.

2

Step 2

Back is available.

3

Step 3

Groups can reduce effort.

Try it

Try: use NEXT until the right group is reached, then SELECT.

You can stop

STOP, WAIT, HELP, NOT THAT and BREAK remain available on every page.

Eye-gaze route

Looking can move focus. A separate dwell, blink, switch or partner confirmation can select.

1

Step 1

A rest zone prevents accidental choice.

2

Step 2

Dwell time is adjustable.

3

Step 3

Tracking loss pauses safely.

Try it

Try: look at a tile, move to REST, then confirm only when ready.

You can stop

STOP, WAIT, HELP, NOT THAT and BREAK remain available on every page.

Head-pointing and dwell

A pointer may follow head movement. Large targets, smoothing and a steady rest area reduce effort.

1

Step 1

No tiny targets.

2

Step 2

No repeated edge crossing.

3

Step 3

Selection timing is adjustable.

Try it

Try: point at a group first, then choose inside the group.

You can stop

STOP, WAIT, HELP, NOT THAT and BREAK remain available on every page.

Keyboard or controller route

Every essential action has a named, remappable control. Menus and gameplay use the same logic.

1

Step 1

No hidden mouse-only step.

2

Step 2

No required simultaneous buttons.

3

Step 3

Toggle alternatives exist.

Try it

Try: remap PREVIEW and COMMIT, then test both.

You can stop

STOP, WAIT, HELP, NOT THAT and BREAK remain available on every page.

A partner names; I decide

A partner may point, describe or scan. The child confirms the choice.

1

Step 1

The partner names choices in order.

2

Step 2

The partner waits.

3

Step 3

The partner repeats the child signal before acting.

Try it

Try: the partner asks, "Did you choose MATCH?" The child confirms YES, NO or NOT SURE.

You can stop

STOP, WAIT, HELP, NOT THAT and BREAK remain available on every page.

Learn through touch

Tactile shape, edge, texture, position and object cues can carry the same rule meaning.

1

Step 1

Touch is offered, not forced.

2

Step 2

Hands explore from a stable home position.

3

Step 3

A spoken or signed summary can accompany touch.

Try it

Try: find a SUPPORT tile by its raised S and dashed edge.

You can stop

STOP, WAIT, HELP, NOT THAT and BREAK remain available on every page.

Learn through listening

A spoken summary names the room, focus, relationship, available actions and result.

1

Step 1

Important sounds have words.

2

Step 2

Speech can be repeated or slowed.

3

Step 3

The child can request concise or detailed mode.

Try it

Try: ask "Where am I?" and "What can I do?"

You can stop

STOP, WAIT, HELP, NOT THAT and BREAK remain available on every page.

Screen-reader play

The screen reader receives objects and relationships, not a vague picture description.

1

Step 1

Stable names.

2

Step 2

State and dependencies.

3

Step 3

Actions and first failed check.

Try it

Try: navigate from a rule to what it needs, then back.

You can stop

STOP, WAIT, HELP, NOT THAT and BREAK remain available on every page.

Sign, gesture or movement

Communication can use sign language, gesture, facial expression or a known body movement.

1

Step 1

The signal is agreed with the child.

2

Step 2

It is not guessed from stereotypes.

3

Step 3

Uncertainty triggers confirmation.

Try it

Try: choose a signal for AGAIN and a different signal for FINISHED.

You can stop

STOP, WAIT, HELP, NOT THAT and BREAK remain available on every page.

A safe mistake

A candidate world lets you try a rule change before it becomes real.

1

Step 1

Preview shows likely result.

2

Step 2

Nothing is committed yet.

3

Step 3

The first unmet check is explained.

Try it

Try: make an intentionally incompatible connection and ask why it is “not yet.”

You can stop

STOP, WAIT, HELP, NOT THAT and BREAK remain available on every page.

Candidate first, real second

The game clearly separates TRY from REAL.

1

Step 1

Candidate state has a clear label and border pattern.

2

Step 2

Commit requires confirmation.

3

Step 3

Cancel returns to the unchanged world.

Try it

Try: compare the candidate and real-world summaries.

You can stop

STOP, WAIT, HELP, NOT THAT and BREAK remain available on every page.

Tell me the first failed check

A rejection should say where the path stopped.

1

Step 1

Outside support?

2

Step 2

Not compatible?

3

Step 3

Guard not reached or evidence uncertain?

Try it

Try: ask for the first failed check, then request one repair option.

You can stop

STOP, WAIT, HELP, NOT THAT and BREAK remain available on every page.

Undo and repair

Repair is part of learning. It should be easy to find and safe to use.

1

Step 1

Undo the last candidate edit.

2

Step 2

Restore the last committed checkpoint when allowed.

3

Step 3

Change the access route without undoing the puzzle.

Try it

Try: make a change, undo it, and replay what happened.

You can stop

STOP, WAIT, HELP, NOT THAT and BREAK remain available on every page.

Take a break

A break pauses time pressure, scanning and sensory output.

1

Step 1

Focus is saved.

2

Step 2

The candidate choice is saved.

3

Step 3

The return page says what is waiting.

Try it

Try: take a break before commit and return to the same candidate.

You can stop

STOP, WAIT, HELP, NOT THAT and BREAK remain available on every page.

Return without a memory test

The game summarizes where you were and what choices are available.

1

Step 1

No "Do you remember?" gate.

2

Step 2

Replay can show the last change.

3

Step 3

The child chooses continue, review or leave.

Try it

Try: use the return card: "You were previewing the MATCH tile."

You can stop

STOP, WAIT, HELP, NOT THAT and BREAK remain available on every page.

My Access Passport

The child may record useful routes, signals and stop preferences without naming a diagnosis.

1

Step 1

What helps.

2

Step 2

What to avoid.

3

Step 3

How I say YES, NO, STOP and NOT SURE.

Try it

Try: fill only the parts the child chooses to share.

You can stop

STOP, WAIT, HELP, NOT THAT and BREAK remain available on every page.

Celebrate ideas, not speed

Success is understanding, choosing, explaining, repairing or collaborating - not being fastest.

1

Step 1

Access tools do not reduce reward.

2

Step 2

Hints do not shame.

3

Step 3

Co-play can earn the same ending.

Try it

Try: praise a clear choice or repair, not quickness.

You can stop

STOP, WAIT, HELP, NOT THAT and BREAK remain available on every page.

Communication is authority

A game action begins with the child's expressed intent. Communication access therefore belongs inside the authoritative input path, not in a side note.

1

Accept many forms

Speech, AAC, sign, tactile sign, object cue, eye gaze, switch, facial expression and individualized movement can all carry intent.

2

Confirm meaning

The system or partner repeats the interpreted choice before commit.

3

Preserve repair

NOT THAT and CHANGED MIND remain available after every interpretation.

System rule

The proposal record stores the selected action and confirmation status, not a diagnosis or judgement about communication ability.

Communication before symbols

Some children communicate before or without conventional symbols. Orientation, approach, withdrawal, stilling, vocalization, breathing change or repeated movement may carry preference.

1

Observe over time

A single movement is not automatically assigned a meaning.

2

Offer real contrasts

Present two genuinely different, safe outcomes.

3

Check consistency

Confirm with people who know the child and remain open to change.

System rule

Pre-symbolic communication requires careful supported decision-making. It must not become mind reading.

Child-specific YES and NO

YES and NO may look different for each child and may change with posture, fatigue or context.

1

Define together

Record the child's current reliable signals.

2

Test neutrally

Use known-answer questions only to verify the route, not to interrogate the child.

3

Allow uncertainty

NOT SURE and NO RESPONSE stay distinct.

System rule

A communication passport may store a description of signals with the child's permission.

No response is not consent

Silence, stillness, delay, looking away, sleep, shutdown or tracking loss must not be treated as YES.

1

Pause

Stop the pending action.

2

Check access

Ask whether the child can perceive and operate the route.

3

Offer choices

WAIT, BREAK, DIFFERENT ROUTE or FINISHED.

System rule

No-response state is explicit in the input model and cannot auto-commit.

AAC core vocabulary

Core words let a child control the process across many rooms.

1

Control

STOP, GO, WAIT, AGAIN,
HELP, FINISHED.

2

Decision

YES, NO, NOT SURE, NOT
THAT, CHANGED MIND.

3

Access

QUIET, DIM, STILL,
BIGGER, READ, TOUCH,
SWITCH ROUTE.

System rule

Core words remain in stable locations so motor and visual learning can develop.

AAC fringe vocabulary

Puzzle-specific words sit beside the core without replacing it.

1

Rule words

SUPPORT, MATCH,
GUARD, PREVIEW,
COMMIT, REPLAY.

2

Object words

TILE, CABLE, GATE,
TRACK, SHEET, SEED.

3

Explanation words

WHERE, WHY, FIRST
FAILED CHECK, WHAT
CHANGED.

System rule

Fringe words use the same stable semantic IDs as onscreen rule objects.

Aided language modeling

Adults may point to or activate AAC words while speaking, without demanding imitation.

1

Model naturally

Use the board during real choices.

2

Do not test

Avoid “show me” drills during a decision.

3

Honor output

When the child uses a word, respond to its meaning.

System rule

Modeling supports access; it does not authorize an adult to select on the child's behalf.

Partner-assisted scanning

A partner presents choices in a predictable order while the child signals when to stop or select.

1

Agree signals

Define YES, NO, STOP and AGAIN.

2

Name neutrally

Use the same wording and pace each pass.

3

Confirm

Repeat the selected item before acting.

System rule

The scan order, group structure and confirmation result can be logged for replay and quality review.

Communication-partner fidelity

The partner is a relay, not a hidden controller.

1

Choice fidelity

Did the committed action match the child's confirmed intent?

2

Pace fidelity

Did the partner wait long enough?

3

Repair fidelity

Was NOT THAT honored immediately?

System rule

Partner fidelity is a release and training metric, not a judgement of the child.

Interpretation repair

Misunderstandings are expected and repairable.

1

Pause

Do not continue from an uncertain interpretation.

2

State the guess

"I thought you chose GUARD."

3

Offer correction

YES, NO, NOT THAT, SHOW AGAIN or DIFFERENT ROUTE.

System rule

The world changes only after repaired confirmation.

Touch consent

Touch support begins with permission and stops at the child's stop signal.

1

Tell before touch

Name who, where and why.

2

Offer alternatives

Object cue, verbal direction, model hand or hand-under-hand.

3

Stop immediately

Withdrawal or STOP ends contact.

System rule

Surprise touch and forced hand movement are prohibited.

Hand-under-hand exploration

The adult's hand acts as a platform or model beneath the child's hand, allowing the child to join, leave or lead.

1

Invite

Offer the object and position.

2

Follow

Let the child control pressure and duration.

3

Describe

Name the object and relation without forcing a response.

System rule

This is “do with,” not “do for.” It requires individualized specialist guidance when used in deafblind education.

Object cues

A real object or consistent miniature can announce an activity before it starts.

1

Stable meaning

The same cue always means the same activity.

2

Tactile access

Shape and texture are distinguishable.

3

Choice use

Offer object cues for PLAY, BREAK and FINISHED.

System rule

Object cues may connect to the same stable activity IDs used digitally.

Tactile schedules

A left-to-right or top-to-bottom sequence of objects shows FIRST, THEN, NEXT and FINISHED.

1

Explore first

Let the child inspect the schedule.

2

Move the marker

The child or partner marks completed steps.

3

Allow change

A changed plan is explained and physically updated.

System rule

The schedule is a prediction aid, not a behavior-compliance contract.

Sign language and interpreters

Signed communication receives equal time, sightlines and authority.

1

Layout

Keep the signer, interpreter and relevant object visible.

2

Timing

Pause visual action while critical signed explanation occurs.

3

Terminology

Agree signs for substrate terms and preserve them consistently.

System rule

Captions do not replace sign language for every child; sign and text can be offered together.

Multilingual communication

The child may understand rules best in a home language, sign language, symbol system or mixed mode.

1

Separate stable ID from label

Translations change the displayed phrase, not the rule identity.

2

Allow text expansion

Layouts reflow for longer words.

3

Check meaning

Use community review for key concepts and safety words.

System rule

The system must never infer lower ability from language choice.

Communication fatigue

Expressing a choice can require more energy than understanding it.

1

Reduce repetition

Do not make the child restate a confirmed choice.

2

Offer shortcuts

Use stored phrases and recent-choice history with permission.

3

Pause scanning

Allow rest without losing focus.

System rule

Effort is measured separately from correctness.

Group communication

Cooperative play needs turn access that is visible and negotiable.

1

Role cards

Observer, Matcher, Guard Keeper, Committer and Replay Guide.

2

Pass or pause

A child may pass without losing status.

3

Shared confirmation

Group choices show whose confirmation is still needed.

System rule

No role is reserved for speaking, reading or physically moving pieces.

Privacy and dignity

Communication preferences can be private and shared only as needed.

1

No public diagnosis label

Show a route, not a medical category.

2

No spectacle

Do not announce that access tools are “special.”

3

Child control

Let the child choose what appears in the Access Passport.

System rule

Access data stays separate from score, behavior ratings and advertising profiles.

Emergency communication

STOP, PAIN, HELP, TOO MUCH, NEED SPACE and CONTACT MY PERSON must remain reachable even when ordinary play is paused.

1

Always available

Do not bury safety words in a submenu.

2

Multiple channels

Provide visible, spoken, tactile and switch-accessible routes.

3

Human response

Define who receives the signal and what they do next.

System rule

Emergency communication does not depend on completing the current scan or puzzle step.

Design for functions, not labels

A diagnosis never predicts one fixed route. Ask what the child needs to perceive, understand, act, confirm, repair, regulate and return.

1

Requirement 1

Offer route combinations.

2

Requirement 2

Let settings change during the session.

3

Requirement 3

Test with the child, not on assumptions.

Parity test

Record supports and barriers, not a deficit score.

Blind access

Build a navigable semantic graph with concise and detailed scene summaries, landmarks, relations, actions and change announcements.

**1**

Requirement 1

No coordinate-only instructions.

**2**

Requirement 2

No visual-only preview.

**3**

Requirement 3

Replay names objects and relations.

Parity test

A blind player can inspect a rule, its dependencies, compatibility and candidate result without using a pixel map.

Low-vision access

Support large text, reflow, magnification, high contrast, strong focus, uncluttered views and stable target positions.

1

Requirement 1

Do not force zoom-out.

2

Requirement 2

Keep text selectable.

3

Requirement 3

Preserve contrast in all states.

Parity test

Magnification does not remove access to commit, undo or error explanation.

Cerebral visual impairment (CVI)

Provide low-complexity views, controlled movement, strong object separation, familiar anchors, extra visual latency and nonvisual equivalents.

1

Requirement 1

One object at a time.

2

Requirement 2

Reduce background detail.

3

Requirement 3

Allow preferred visual field and viewing distance.

Parity test

CVI support is individualized; the system offers adjustable visual complexity rather than a single "CVI mode."

Color-vision access

Color can reinforce meaning but never carry it alone.

1

Requirement 1

Add word labels.

2

Requirement 2

Add shape and border pattern.

3

Requirement 3

Add position or tactile code.

Parity test

Every state difference remains identifiable in monochrome.

Deaf access

Provide captions, sign-compatible layouts, visible timing, text transcripts and visual equivalents for all audio cues.

1

Requirement 1

Do not cover the signer with UI.

2

Requirement 2

Pause competing visual action during key signed information.

3

Requirement 3

Let captions name speaker and sound source.

Parity test

Rhythm and timing cues have visible or tactile forms.

Hard-of-hearing access

Allow independent speech, music and effect levels; clear captions; replay; and optional hearing-device-friendly audio settings.

1

Requirement 1

No whisper-only clue.

2

Requirement 2

No alert hidden in music.

3

Requirement 3

Repeat without penalty.

Parity test

Audio is supportive, never the only route to rule state.

Deafblind access

Use stable tactile orientation, object cues, tactile symbols, hand-under-hand exploration, tactile schedules and partner-confirmed communication.

1

Requirement 1

Tell before touch.

2

Requirement 2

Keep a tactile home position.

3

Requirement 3

Represent change through object movement or replay beads.

Parity test

Every critical tactile interaction follows consent and individualized communication practice.

Motor access

Avoid precision, sustained force, rapid repetition and mandatory simultaneous actions.

1

Requirement 1

Large targets.

2

Requirement 2

Adjustable snap and dwell.

3

Requirement 3

Toggle, step and partner-assisted alternatives.

Parity test

Motor effort is not part of puzzle difficulty unless explicitly chosen by the child.

One-hand access

Every two-hand action has a sequential or toggle route.

1

Requirement 1

Anchor then transform.

2

Requirement 2

Lock one axis at a time.

3

Requirement 3

Use mode cycling instead of chords.

Parity test

The conceptual relation remains the same even when bimanual input is serialized.

Switch access

Provide one-switch auto-scan, two-switch next/select, step scan, grouping, back, pause and adjustable timing.

1**Requirement 1**

Stable order.

2**Requirement 2**

Visible and spoken focus.

3**Requirement 3**

No timeout after selection.

Parity test

Switch scanning reaches every essential action, including settings, help, undo and exit.

Gaze and head access

Use large stable targets, rest zones, adjustable dwell, smoothing, confirmation and safe tracking-loss behavior.

**1**

Requirement 1

No gaze-only irreversible commit.

**2**

Requirement 2

Avoid edge-crowded targets.

**3**

Requirement 3

Keep a nonmoving rest area.

Parity test

Tracking loss pauses and preserves focus instead of interpreting movement as a choice.

Tremor, ataxia and dyspraxia

Support movement filtering, target expansion, snap, undo, delayed activation and non-drag routes.

1

Requirement 1

Do not punish repeated entries.

2

Requirement 2

Offer discrete direction controls.

3

Requirement 3

Separate hover from select.

Parity test

Accidental activation is repairable without score loss.

Speech, AAC and nonspeaking access

No essential action, explanation or social role requires speech.

1

Requirement 1

Named actions map to AAC.

2

Requirement 2

Text entry has symbol/choice alternatives.

3

Requirement 3

Voice recognition is optional.

Parity test

Communication route does not alter reward or story status.

Cognitive and intellectual disability access

Use concrete language, stable routines, visible cause and effect, one decision at a time and meaningful repetition.

1

Requirement 1

Show before asking.

2

Requirement 2

Use real examples.

3

Requirement 3

Avoid trick wording.

Parity test

The formal concept is preserved while explanation depth and pacing are adjustable.

Reading and dyslexia access

Offer narration, plain language, short lines, generous spacing, symbol support and a no-reading route.

1

Requirement 1

Do not use all caps for paragraphs.

2

Requirement 2

Avoid dense justified text.

3

Requirement 3

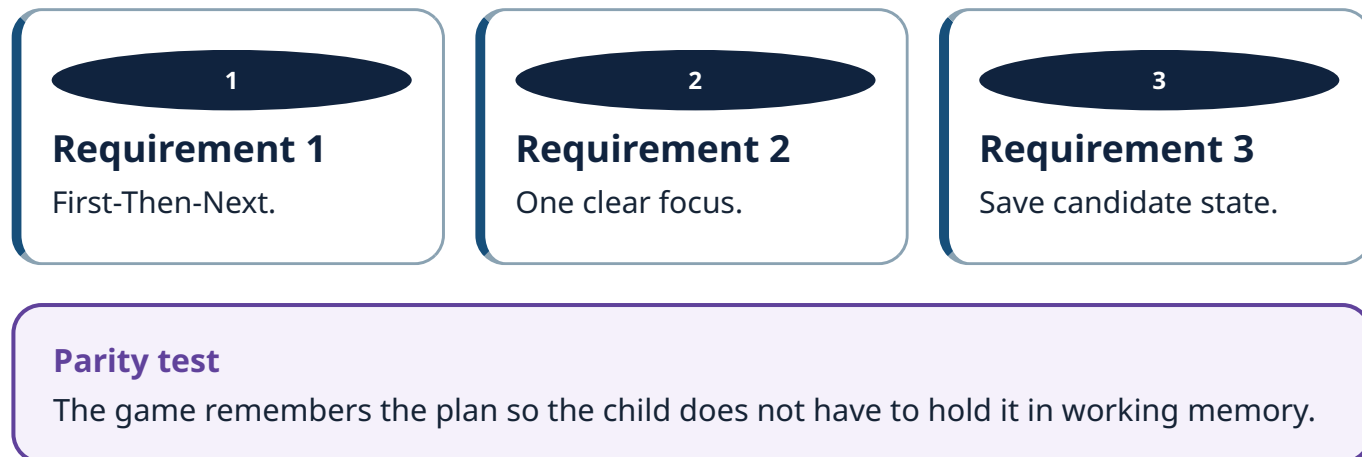
Keep words selectable and reflowable.

Parity test

Reading skill is not a gate to topology or rule reasoning.

Executive-function access

Externalize sequence, status, next action, unfinished choices and return point.



Autistic and sensory access

Provide predictability, direct language, sensory controls, interests as optional bridges, and no forced eye contact or social performance.

1

Requirement 1

Preview transitions.

2

Requirement 2

Offer quiet/still routes.

3

Requirement 3

Allow solo or structured co-play.

Parity test

Regulation is access, not a reward for compliance.

Attention variability

Use short chunks, resumable goals, persistent focus, low-distraction mode and optional movement breaks.

1

Requirement 1

No rapid reminder spam.

2

Requirement 2

Let the child mark “come back later.”

3

Requirement 3

Keep important state visible.

Parity test

Attention changes do not erase understanding or decision authority.

Photosensitivity and migraine

Avoid flashing, high-contrast flicker, rapid luminance change and unavoidable particle effects; provide a low-stimulation preset before first play.

1

Requirement 1

No flashing essential clue.

2

Requirement 2

Motion and glow can be disabled.

3

Requirement 3

Pause immediately.

Parity test

Safety settings are available before any potentially triggering content.

Vestibular and motion sensitivity

Offer static camera, reduced parallax, no camera shake, slower transitions and nonanimated alternatives.

1

Requirement 1

No forced head motion.

2

Requirement 2

No zoom pulse.

3

Requirement 3

Provide teleport or step transitions where relevant.

Parity test

Reduced motion retains event timing and state information.

Fatigue, pain and chronic illness

Support micro-sessions, save anywhere, low-force input, effort forecasts, quiet pauses and return summaries.

1

Requirement 1
No streak loss.

2

Requirement 2
No repeated calibration.

3

Requirement 3
No access setup tax each session.

Parity test
Energy varies; the child can change route without losing the world.

Trauma-informed and emotional safety

Use clear boundaries, predictable consequences, opt-out routes, nonpunitive repair and no forced disclosure.

1

Requirement 1

Avoid surprise restraint or touch.

2

Requirement 2

Do not simulate helplessness as onboarding.

3

Requirement 3

Keep exit and pause visible.

Parity test

The game never uses access needs as horror, punishment or moral failure.

Combined access

Test routes in real combinations: screen reader plus switch, low vision plus gaze, AAC plus partner scan, deafblind plus fatigue.

1

Requirement 1

Features must coexist.

2

Requirement 2

Focus and captions must not conflict.

3

Requirement 3

One route must not disable another.

Parity test

Single-feature compliance is insufficient when children use combined access.

Fluctuating access and device loss

Access may change within minutes. A device disconnect, tracking loss or overload must pause safely and preserve the same object, candidate and confirmation state.

1

Requirement 1

Offer another route.

2

Requirement 2

Explain what was preserved.

3

Requirement 3

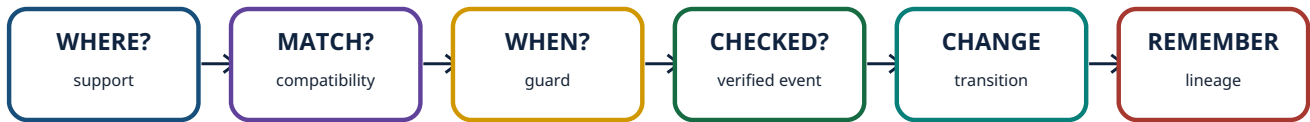
Resume without recalibration when possible.

Parity test

Temporary access session state is saved separately from authoritative puzzle state.

A world made of rules

Definition Breaker turns invisible rules into objects that can be noticed, moved, connected, checked and remembered.



Child words

Definition Breaker turns invisible rules into objects that can be noticed, moved, connected, checked and remembered.

Grown-up bridge

A door can carry its own support, compatibility and guard cards. A child can inspect the reason it opens instead of memorizing a hidden script.

Formal substrate

Definitions, instances, relations, supports, compatibility gates, events, transitions and lineage are first-class substrate objects.

Play prompt

Ask the child to show the idea with a tile, word, sound, gesture, switch or partner-assisted choice.

Why questions came first

The substrate began by asking what needs to be known before a change is allowed.

Child words

The substrate began by asking what needs to be known before a change is allowed.

Grown-up bridge

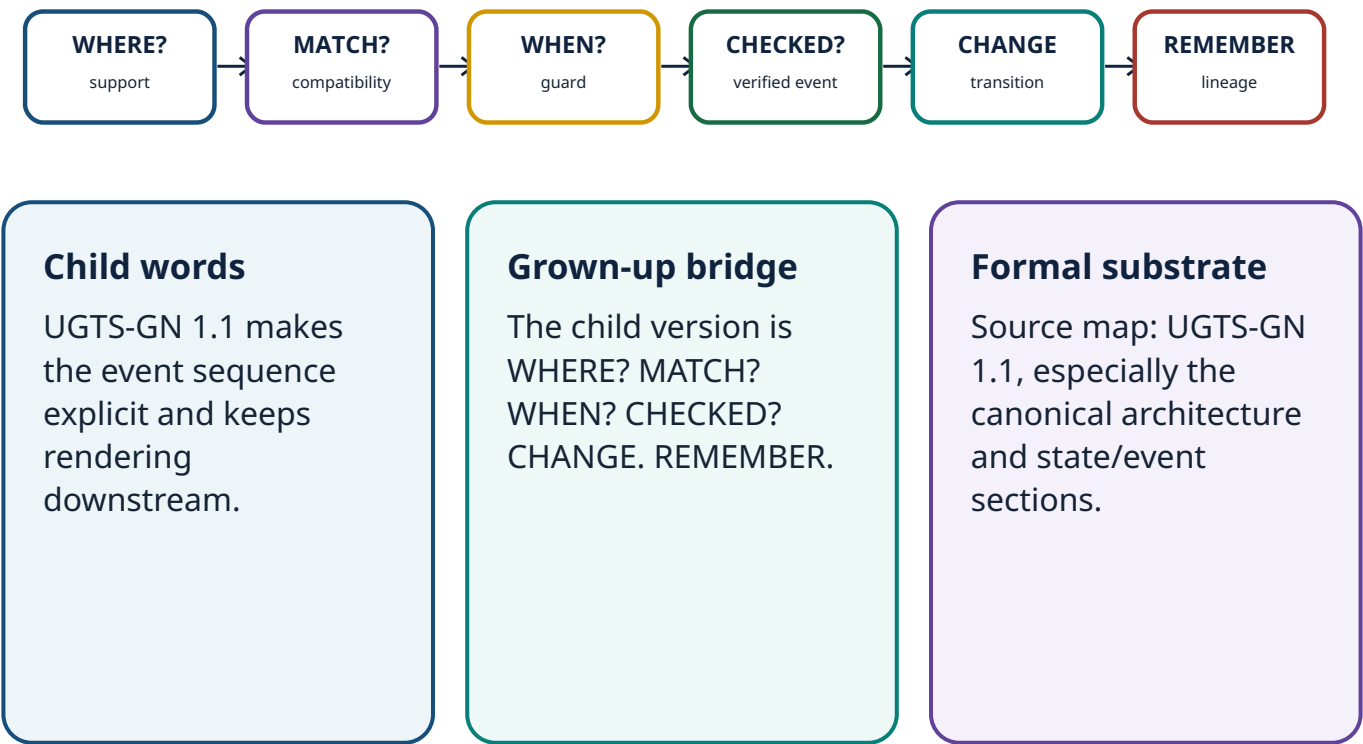
First find what is nearby and relevant. Then ask whether it may interact. Then check a boundary. Only then may the world change.

Formal substrate

The canonical query-first authority chain is support -> compatibility -> guard -> verified event -> transition -> lineage.

GN 1.1: the six-question backbone

UGTS-GN 1.1 makes the event sequence explicit and keeps rendering downstream.



KC 2.0: patterns, fields and motion

KC 2.0 adds shape families, implicit fields, route identity, moving frames, bounded dynamics and hybrid events.



Child words

KC 2.0 adds shape families, implicit fields, route identity, moving frames, bounded dynamics and hybrid events.

Grown-up bridge

Children can meet these as path cards, field mats, motion arrows and “same place, different journey” stories.

Formal substrate

Patterns and dynamics extend the operator set but do not replace support, compatibility, guards or lineage.

Two Hands 3.0: interaction, proposals and replay

Two Hands 3.0 adds stable scene nodes, hand interaction, candidate proposals, deterministic commit, checkpoints and replay.

Child words

Two Hands 3.0 adds stable scene nodes, hand interaction, candidate proposals, deterministic commit, checkpoints and replay.

Grown-up bridge

A child may grab a rule tile, preview the effect, confirm, and replay the accepted change.

Formal substrate

Input systems may propose in parallel; only verified deterministic commit changes authoritative state.

KC 3.6: definitions live inside the world

KC 3.6 turns definitions, dependencies, phases and operation order into inspectable records.

Child words

KC 3.6 turns definitions, dependencies, phases and operation order into inspectable records.

Grown-up bridge

A rule card can say what it is, what it needs, when it runs and which invariant it must keep.

Formal substrate

Each literal definition has stable ID, type, dependencies, phase, invariants, provenance and content hash.

SCLP 3.6.2: bounded worlds and winding

SCLP adds guard intervals, phase plus winding, reflective topology, constraint release, bounded grammar and finite key layouts.

Child words

SCLP adds guard intervals, phase plus winding, reflective topology, constraint release, bounded grammar and finite key layouts.

Grown-up bridge

Children can count loops, flip a track, remove one shackle, grow a finite branching plant and compare seed cards.

Formal substrate

The profile is finite and query-first; compression claims require reconstruction and error equivalence.

KC 3.9: a playable vector runtime

KC 3.9 adds vector assets, named multi-device actions, fixed-step simulation, collisions, animation, sound, persistence and offline HTML5.

Child words

KC 3.9 adds vector assets, named multi-device actions, fixed-step simulation, collisions, animation, sound, persistence and offline HTML5.

Grown-up bridge

The accessible game can use the same project data for keyboard, gamepad, touch, switch adapters and deterministic replay.

Formal substrate

One versioned GameProject can validate, simulate and build to self-contained browser output.

SARA 3.6.3: public, private and permitted

For older learners, SARA shows that typed information can be public, private or permitted only under an authorization guard.

Child words

For older learners, SARA shows that typed information can be public, private or permitted only under an authorization guard.

Grown-up bridge

A public label does not give permission to discover or use a secret. Some paths must be rejected.

Formal substrate

This is an optional ethics and security lesson. Secret search, signing, broadcast and funds transfer stay outside children's play.

The six big questions

Every rule change can be explained through six short questions.



Child words

Every rule change can be explained through six short questions.

Grown-up bridge

WHERE can it work?
MATCH may these things work together?
WHEN did the boundary happen?
CHECKED is the evidence good enough?
CHANGE what becomes different?
REMEMBER what must be kept?

Formal substrate

These are the child-language bridge to support, compatibility, guard, verified event, transition and lineage.

Play prompt

Ask the child to show the idea with a tile, word, sound, gesture, switch or partner-assisted choice.

Definitions

A definition says what kind of rule something is and how it behaves.

Child words

A definition says what kind of rule something is and how it behaves.

Grown-up bridge

A SUPPORT tile might define a circle around a sensor. Every tile with that definition can use the same rule.

Formal substrate

A literal definition record declares ID, kind, domain, codomain, parameters, dependencies, phase, invariants and provenance.

Stable identity

A thing can move or change appearance without becoming a different thing.

Child words

A thing can move or change appearance without becoming a different thing.

Grown-up bridge

The blue bridge tile keeps its identity when the player rotates it or changes its material.

Formal substrate

Node identity, asset identity, coordinates and lineage are distinct.

Dependencies

A dependency says what a rule needs before it can work.

Child words

A dependency says what a rule needs before it can work.

Grown-up bridge

The OPEN rule may need a KEY rule and a MATCH rule. Pulling the OPEN card also shows those needed cards.

Formal substrate

Definition dependencies form an acyclic graph and are resolved before evaluation.

Operation order

Doing the same actions in a different order can make a different world.

Child words

Doing the same actions in a different order can make a different world.

Grown-up bridge

MOVE then ROTATE is not always the same as ROTATE then MOVE. The order rail makes this visible.

Formal substrate

Composition is generally noncommutative; the evaluation schedule is explicit and traceable.

Hinges

A hinge connects, pivots or selects a map. It can matter structurally even when it adds no number.

Child words

A hinge connects, pivots or selects a map. It can matter structurally even when it adds no number.

Grown-up bridge

A connector tile can join UNIT, EN and TENS. A parity hinge can choose route A or B.

Formal substrate

A hinge carries pivot or connector locus, state, selected map, guard and invariants.

Support

Support asks whether a rule is relevant here.

Child words

Support asks whether a rule is relevant here.

Grown-up bridge

A sound gate may work only inside the blue circle. Outside the circle, it is not considered.

Formal substrate

Local spatial support prunes irrelevant candidates before expensive relation work.

Compatibility

Compatibility asks whether supported things are allowed to couple.

Child words

Compatibility asks whether supported things are allowed to couple.

Grown-up bridge

Two tokens may share a square but belong to different sheets, teams or phases, so they do not interact.

Formal substrate

Compatibility uses sheet, phase, orientation, mode, policy, ownership and application tags.

Guards

A guard watches for a declared boundary or condition.

Child words

A guard watches for a declared boundary or condition.

Grown-up bridge

A door guard may wait until the counter reaches three. Near the boundary, it can report NOT SURE rather than guess.

Formal substrate

Guard classification uses a relation, margin, uncertainty and crossing/touch/tangency policy.

Candidate world

A candidate world is a safe “what if?” state.

Child words

A candidate world is a safe “what if?” state.

Grown-up bridge

The child connects a new rule and sees the predicted route without changing the real board.

Formal substrate

Candidate proposals remain nonauthoritative until support, compatibility, guard and policy checks pass.

Verified proposal

A proposal becomes verified only when every required check passes with enough confidence and small enough error.

Child words

A proposal becomes verified only when every required check passes with enough confidence and small enough error.

Grown-up bridge

The game can say: "It is in range, it matches, the guard crossed, and the evidence is clear."

Formal substrate

verified = support_ok
and compatibility_ok
and accepted_guard
and confidence >=
floor and
numeric_error <=
margin.

Commit and change

Commit applies one checked patch to the real world.

Child words

Commit applies one checked patch to the real world.

Grown-up bridge

The candidate bridge becomes the real bridge only after the child confirms COMMIT.

Formal substrate

Verified proposals are ordered, conflicts are resolved by versioned policy, and one authoritative patch is applied.

Lineage

Lineage remembers where an object or rule came from and how it changed.

Child words

Lineage remembers where an object or rule came from and how it changed.

Grown-up bridge

A split token becomes child A and child B. A merged object records both parents.

Formal substrate

Durable identity uses generative address, transition history, invariants and novelty records rather than coordinates alone.

Replay

Replay rebuilds a world from a start state, ordered events and checkpoints.

Child words

Replay rebuilds a world from a start state, ordered events and checkpoints.

Grown-up bridge

The child can hear or feel: "You moved MATCH, previewed, committed, then changed the route."

Formal substrate

Each event carries sequence and pre/post hashes; replay stops at the first divergence.

Coordinates are not identity

Two things can be at the same place and still be different.

Child words

Two things can be at the same place and still be different.

Grown-up bridge

A bridge on sheet A and a bridge on sheet B overlap in the picture but need not connect.

Formal substrate

Position equality does not imply coupling or identity; sheet, route, orientation and lineage remain separate.

Sheets and phases

A sheet or phase is an extra state label that separates worlds sharing the same coordinates.

Child words

A sheet or phase is an extra state label that separates worlds sharing the same coordinates.

Grown-up bridge

A red phase token can pass through a blue phase wall until a MATCH rule changes.

Formal substrate

Sheet and phase are typed compatibility fields, not hidden visual effects.

Topology and orientation

Topology describes how places are connected and how orientation changes across a boundary.

Child words

Topology describes how places are connected and how orientation changes across a boundary.

Grown-up bridge

A track can wrap so LEFT becomes RIGHT after one trip. The transition is shown by an arrow and flip token.

Formal substrate

Möbius or Klein behavior is an explicit gluing map with coordinate, sheet, orientation and lineage transport.

Winding remembers loops

Phase can repeat while winding remembers how many full loops happened.

Child words

Phase can repeat while winding remembers how many full loops happened.

Grown-up bridge

Two tokens at the same clock position can have winding 1 and winding 4, so their histories differ.

Formal substrate

Linear time maps to periodic phase plus integer winding lineage.

Patterns

A pattern is a versioned shape or path with parameters and a declared role.

Child words

A pattern is a versioned shape or path with parameters and a declared role.

Grown-up bridge

Rose, spiral, clothoid and Gielis cards can make routes, supports or puzzle boundaries.

Formal substrate

Pattern instances declare mechanism ID, parameters, frame, support, compatibility tags and validation contract.

Fields

A field gives a value at each place. Zero can be a boundary, but not every field is an exact distance.

Child words

A field gives a value at each place. Zero can be a boundary, but not every field is an exact distance.

Grown-up bridge

A heat mat can say negative inside, zero at the edge and positive outside.

Formal substrate

Implicit-field sign, gradient and guard semantics are separate from exact signed-distance capability.

Motion and frames

Motion says where an object is and how its local directions move.

Child words

Motion says where an object is and how its local directions move.

Grown-up bridge

A car on a curved track has forward, side and up directions that change along the path.

Formal substrate

Kinematic queries return `state_at(t)`, `frame`, `curvature` and `bounded status`; a frame loop is not always required.

Constraints

A constraint keeps a relation true over an interval.

Child words

A constraint keeps a relation true over an interval.

Grown-up bridge

A token clipped to a rail can move along it but not leave it.

Formal substrate

Resting contact, joints and cooperative grabs are maintained constraints, not infinite zero-crossing events.

The missing shackle

Removing one independent constraint can add one legal direction of motion.

Child words

Removing one independent constraint can add one legal direction of motion.

Grown-up bridge

Take away one bar from a sliding frame and test the new movement. Do not call the result chaos automatically.

Formal substrate

Constraint-row deletion changes rank and nullity; dynamics require a separate mass, force and damping model.

Bounded grammar and branches

A grammar grows a structure using declared productions and strict budgets.

Child words

A grammar grows a structure using declared productions and strict budgets.

Grown-up bridge

A seed card grows a branching path for four levels, then stops. Every push has a matching pop.

Formal substrate

Depth, symbol count, stack and branch density are finite and overflow is rejected.

Seeds, keys and honest compression

A seed can rebuild regular structure, while true outside changes still need event records.

Child words

A seed can rebuild regular structure, while true outside changes still need event records.

Grown-up bridge

A short world code may reconstruct a maze, but it cannot magically contain infinite detail.

Formal substrate

Finite keys have declared ranges and quantization; width ratios are not equivalent compression without semantic and error equivalence.

Capstone: explain a full rule change

Use the whole path to explain one puzzle solution.

Child words

Use the whole path to explain one puzzle solution.

Grown-up bridge

WHERE: the bridge is in the bay. MATCH: the key and lock agree. WHEN: the counter crossed three. CHECKED: uncertainty is below the margin. CHANGE: the bridge opens. REMEMBER: replay records the accepted rule.

Formal substrate

A complete trace names definitions, dependencies, evaluation order, support, compatibility, guard status, proposal, commit patch and lineage.

Play prompt

Ask the child to show the idea with a tile, word, sound, gesture, switch or partner-assisted choice.

One semantic world

All input and output routes operate on the same semantic objects and authoritative state.

1

One object ID

Visual tile, spoken label, tactile symbol and AAC word refer to the same rule ID.

2

One proposal path

Keyboard, switch, gaze and partner input produce the same named action proposal.

3

One outcome

Verification, reward, story consequence and replay are route-independent.

Implementation contract

Do not fork an “accessible game” with different rules unless the child deliberately chooses a different difficulty.

AccessProfile stays separate

Store access preferences separately from score, achievement, puzzle difficulty and authoritative world state.

1

Private by default

The child chooses what is saved and shared.

2

No diagnosis inference

Settings do not become a guessed medical label.

3

Portable when chosen

A child may export preferences without exporting gameplay history.

Implementation contract

AccessProfile may include presentation, input, communication, language and regulation preferences.

AccessSessionState is temporary

Keep current scan focus, gaze rest state, caption position, communication route and sensory level as resumable session state.

1

Preserve interruption

Pause, device loss and route change keep state.

2

Expire safely

Temporary details can clear when the child chooses.

3

Never score it

Session access state is not performance data.

Implementation contract

Separate session state prevents a switch or gaze route from contaminating puzzle identity.

Named actions

Gameplay consumes stable intent names rather than device-specific buttons.

1

Examples

MOVE, INSPECT,
PREVIEW, COMMIT,
UNDO, STOP, HELP,
BREAK.

2

Multiple bindings

Keyboard, gamepad,
touch, switch, gaze and
partner adapters can
bind to the same action.

3

Full menus

Settings and shell
controls use named
actions too.

Implementation contract

KC 3.9 already uses device-independent action maps; DB-2.0 extends this to access and communication actions.

Input adapters

Adapters translate device signals into named action values and edge events without changing rule semantics.

1

Validate range

Clamp, debounce and report confidence.

2

Expose status

Tracking unavailable and device disconnected are states, not silent failure.

3

Replay intent

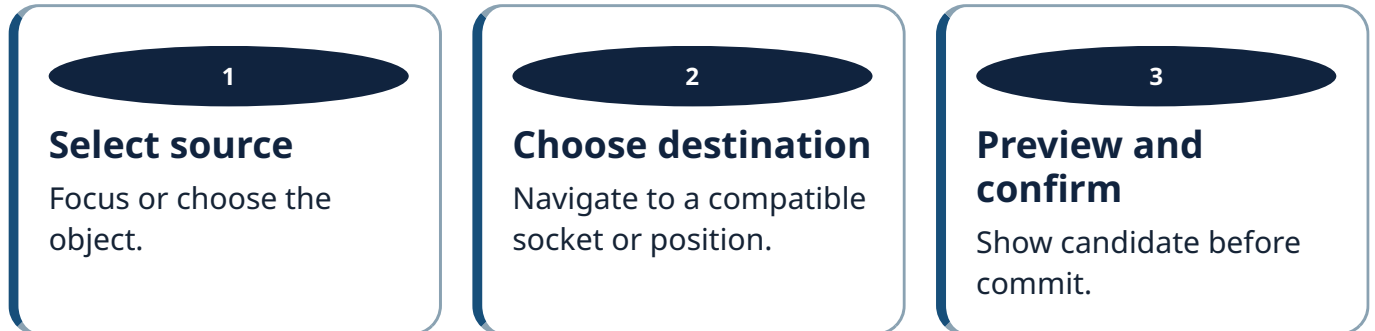
Record resolved actions, not raw biometric streams by default.

Implementation contract

Assistive input remains downstream of child intent and upstream of proposal verification.

No-drag route

Every drag has select-destination, directional step, snap, scan or partner-assisted equivalent.



Implementation contract

The no-drag route reaches the same placement precision through discrete semantics.

No-hold route

Every sustained hold has toggle, timed auto-complete, latch or separate start/stop equivalent.

1

Requirement 1

Hold to inspect -> toggle inspect.

2

Requirement 2

Hold to charge -> choose amount then confirm.

3

Requirement 3

Hold and move -> anchor then move sequentially.

Implementation contract

Holding strength or duration is not required unless explicitly chosen as the challenge.

Remapping and conflict detection

All named actions can be remapped, and the system warns when bindings collide.

1

Accessible before play

Remapping works on first launch.

2

Reset and undo

A safe default is always recoverable.

3

Per context

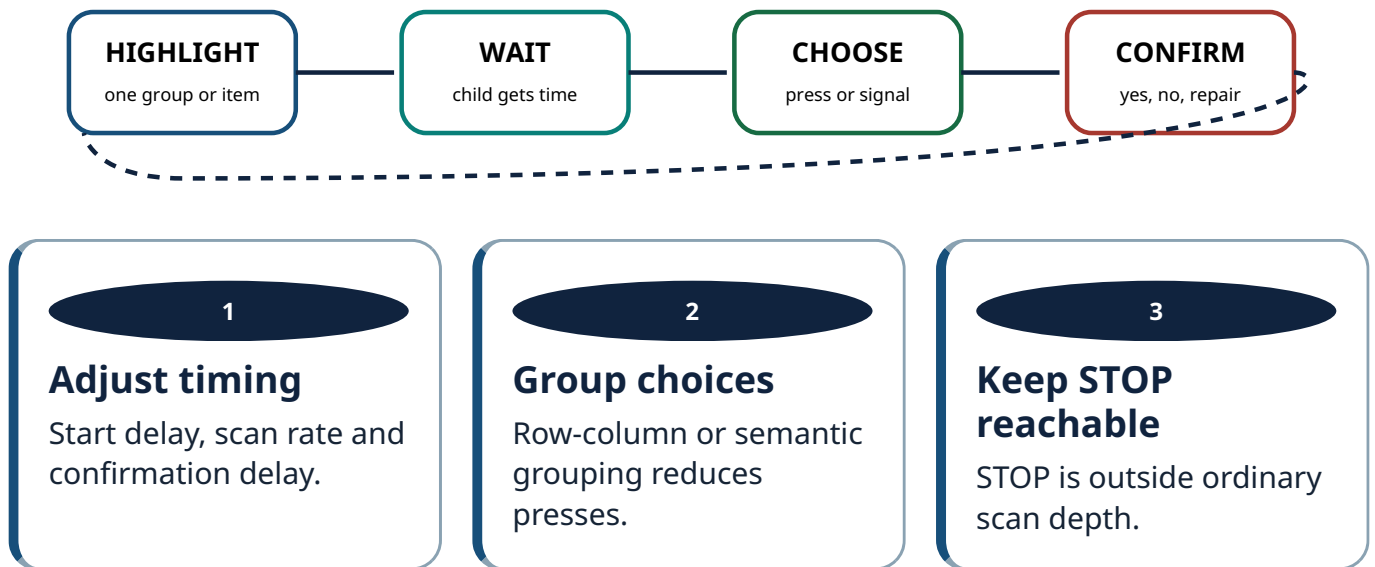
Menu and gameplay conflicts are explained.

Implementation contract

Do not reserve important keys or buttons without an alternative.

Switch scanning state machine

Scanning has explicit highlight, wait, choose, confirm, back and pause states.

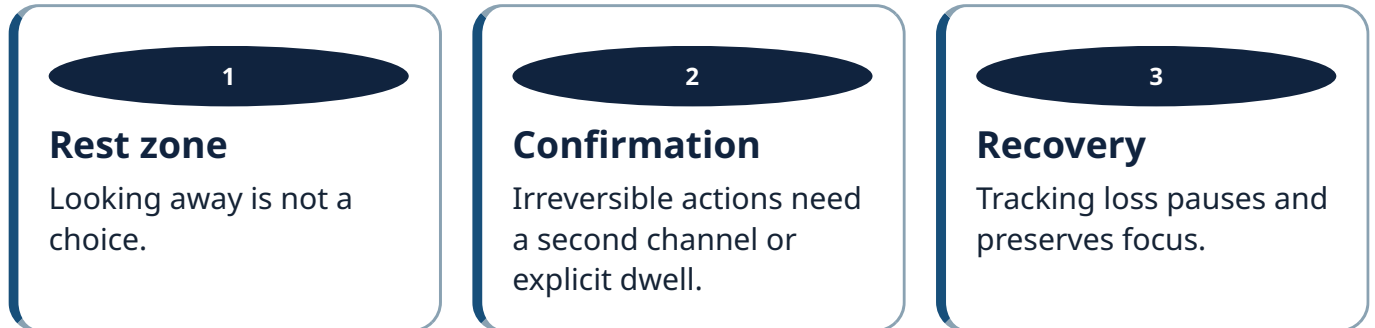


Implementation contract

Scan state survives pause and does not advance while the system is speaking a required label.

Gaze state machine

Gaze navigation separates pointing, dwell, rest, confirmation and tracking-loss states.



Implementation contract

Raw gaze paths are not stored by default.

Predictable focus

Focus moves in a stable logical order and returns to the initiating object after dialogs, help or settings.

1

Visible and spoken

Focus has high-contrast and screen-reader state.

2

No focus trap

Every modal has cancel and return.

3

No surprise reset

Filtering or route change preserves the nearest semantic focus.

Implementation contract

Focus order follows semantic structure, not arbitrary screen coordinates.

Screen-reader object contract

Every rule object exposes name, role, state, value, description, dependencies, relations, available actions and change history.

1 Name "Compatibility seal B."	2 State "Candidate, disconnected, first failed check: phase mismatch."	3 Actions Connect, inspect dependency, preview, cancel.
--	--	---

Implementation contract

The semantic record is authoritative for nonvisual navigation; alt text alone is insufficient.

Queryable graph summaries

Provide commands such as scene summary, current object, neighbors, dependencies, route history, failed check and available repair.

1

Concise mode

Only changed or actionable facts.

2

Detailed mode

Full relation and provenance trace.

3

Landmarks

Bench, rule shelf, candidate mat, exit.

Implementation contract

Queries use stable IDs so replay and focus remain coherent.

Captions and narration

All speech and meaningful audio have captions; essential text can be narrated.

1

Speaker and source

Identify who or what produced the sound.

2

Timing

Keep captions long enough and avoid covering controls.

3

Control

Pause, replay, speed and verbosity are adjustable.

Implementation contract

Narration and captions are synchronized to semantic events rather than only video time.

Sound cues have equivalents

A cue may be sound, visible pulse, caption, haptic pattern or tactile device output - never sound alone.

1 Name the event "Guard crossed."	2 Name urgency Informational, warning or stop.	3 Let the child choose Turn channels on or off independently.
---	--	---

Implementation contract

Procedural audio from KC 3.9 remains presentation, not authority.

Haptics are optional and named

Vibration patterns can reinforce state but must be predictable, adjustable and replaceable.

1

Requirement 1

No surprise vibration.

2

Requirement 2

Intensity and duration are limited.

3

Requirement 3

A visual/spoken/tactile alternative exists.

Implementation contract

Haptic feedback never carries unique safety or puzzle information.

Text and reading

Use plain language, meaningful headings, short paragraphs, selectable text, scalable type and reflow.

1

Child layer

Simple words and examples.

2

Builder layer

More precise concepts.

3

Formal layer

Original substrate terms and contracts.

Implementation contract

Reading level can change without changing object identity or rule consequence.

Reduced motion

Allow static camera, no shake, no parallax, slower transitions, reduced particles and instant state changes.

1

Requirement 1

Show before/after.

2

Requirement 2

Preserve timing in text.

3

Requirement 3

No flashing essential cue.

Implementation contract

Motion setting is available before the first animated scene.

Sensory forecast

Before a room, expose sound, light, motion, touch/haptic, reading, timing and social-demand descriptors.

1

Intensity

none, low, medium, high.

2

Control

adjustable, avoidable or required.

3

Alternative

available route.

Implementation contract

Forecast values are authored and tested, not guessed at runtime from the child.

Constructive error explanations

A rejected action names the first failed rule and an available repair.

1

Not “wrong.”

Say “not yet: outside support.”

2

Show evidence

Name residual or mismatch in suitable language.

3

Offer next step

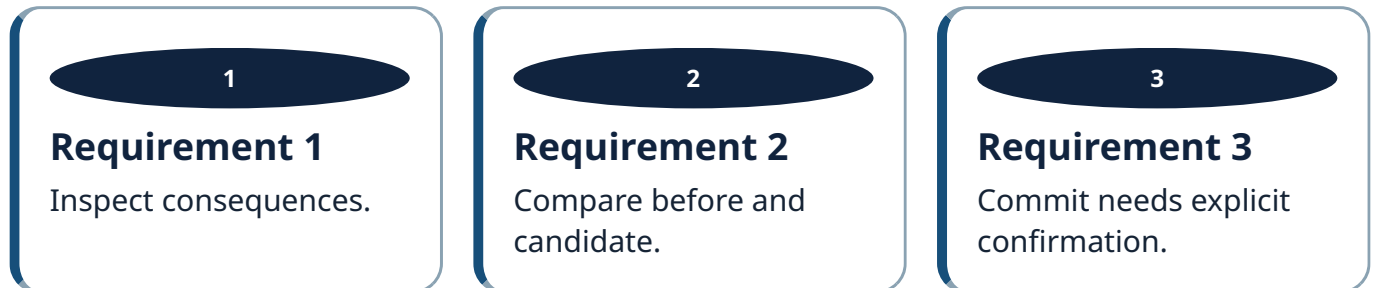
Move closer, change phase, inspect dependency or cancel.

Implementation contract

Error explanation is available through every output route.

Candidate preview

Preview clearly separates proposed and real state through label, border pattern, spoken cue and semantic status.



Implementation contract

Candidate state cannot silently trigger irreversible gameplay or analytics.

Undo and replay

Undo restores a previous valid state where the puzzle allows it; replay explains accepted changes even when undo is not allowed.

**1**

Requirement 1

Named checkpoint.

**2**

Requirement 2

Accessible event
summary.

**3**

Requirement 3

First divergence report
for technical debugging.

Implementation contract

Undo and replay controls are reachable through all access routes.

Save and return

Save focus, candidate, route and communication state without requiring the child to reconstruct context.

1

Return summary.

Continue, review or leave.

2

No streak penalty.

No repeated first-run calibration.

Implementation contract

KC 3.9 persistence can store canonical game state; access preferences remain separately scoped.

Multiplayer and co-play

Accessible routes retain equal authority in local and networked play.

1

Requirement 1

Turn ownership is explicit.

2

Requirement 2

Partner actions identify the child they relay for.

3

Requirement 3

Latency accommodation does not lower reward.

Implementation contract

Authoritative commit records whose confirmed intent produced each proposal, without exposing private access data.

Offline and low-end support

Core access works without a network, cloud speech service or high-end device.

1

Requirement 1

Self-contained build.

2

Requirement 2

Text and captions local.

3

Requirement 3

Reduced effects and low-compute view.

Implementation contract

KC 3.9 provides a self-contained HTML5 direction; cloud-only access is not acceptable for essential play.

Privacy and child data

Collect the minimum needed for function and evidence.

1

Requirement 1

No emotion recognition.

2

Requirement 2

No raw gaze retention by default.

3

Requirement 3

No advertising profile from access settings.

Implementation contract

Children and caregivers can inspect, export or delete stored access information.

Accessibility lint

Authoring tools automatically flag missing semantic names, single-channel clues, drag-only actions, inaccessible focus, missing captions and unequal rewards.

1

Requirement 1

Block severe defects.

2

Requirement 2

Link the exact object and action.

3

Requirement 3

Allow documented review, not silent ignore.

Implementation contract

Lint complements child testing; it does not certify accessibility alone.

Release gates

A build cannot ship when an essential route fails the eight-stage envelope or parity tests.



Implementation contract

Release evidence includes test cases, participant feedback, known limitations, defect owners and retest results.

Physical play principles

Physical play teaches the same concepts through shape, texture, position, connection, order and replay.

1

Same semantic IDs

A physical SUPPORT tile and digital SUPPORT object share a code.

2

Multiple perception routes

Word, letter, shape, border pattern, tactile mark and spoken name.

3

Child control

Pieces, partners and schedules preserve STOP, WAIT, NOT THAT and BREAK.

Prototype boundary

All designs are prototype scaffolds pending child co-design, material review, age grading and toy-safety assessment.

Accessible rule tile anatomy

A rule tile combines redundant identity cues and typed connection points.



letter + word + border pattern + tactile mark + typed socket

1

Face

Large word, letter code and optional symbol.

2

Edge

Distinct border pattern and tactile notch.

3

Socket

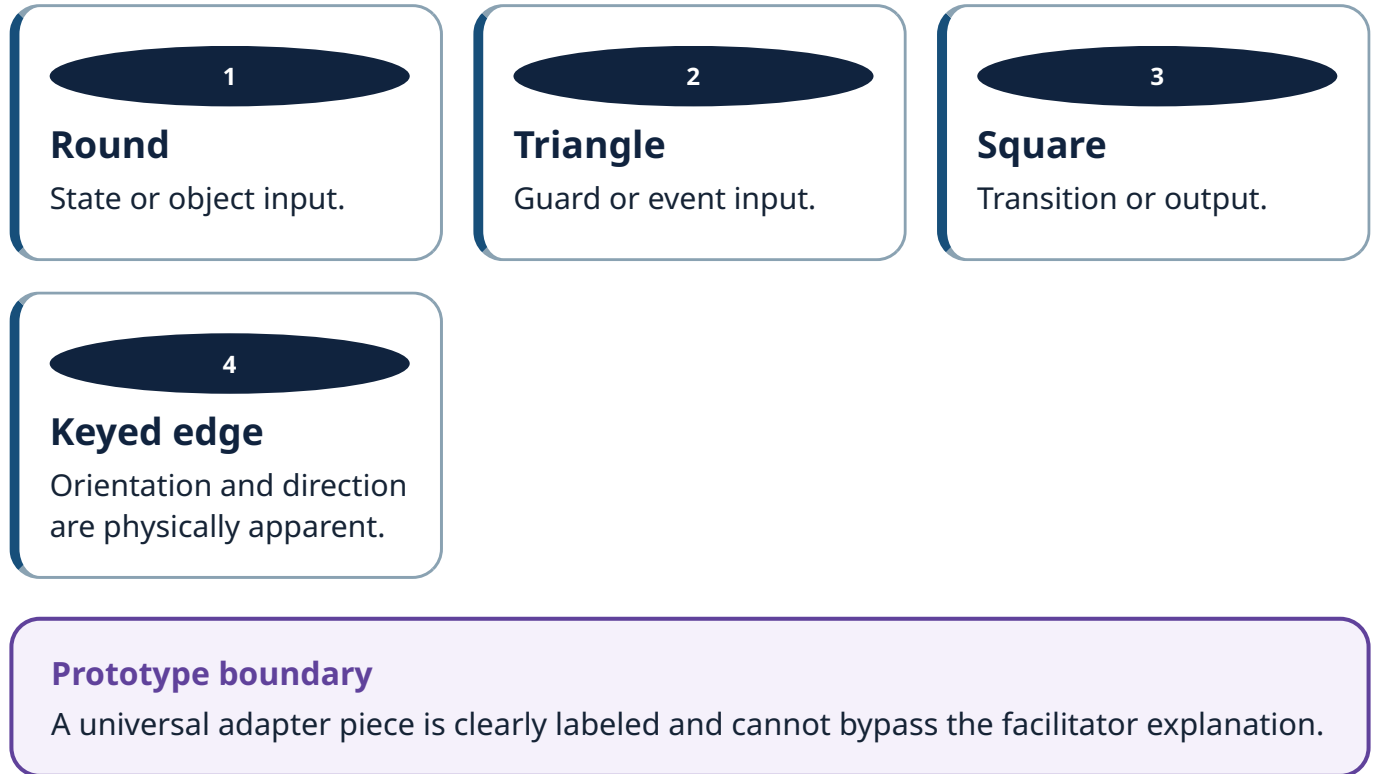
Shape-coded connector that rejects incompatible links mechanically where possible.

Prototype boundary

No essential distinction relies on color or tiny print.

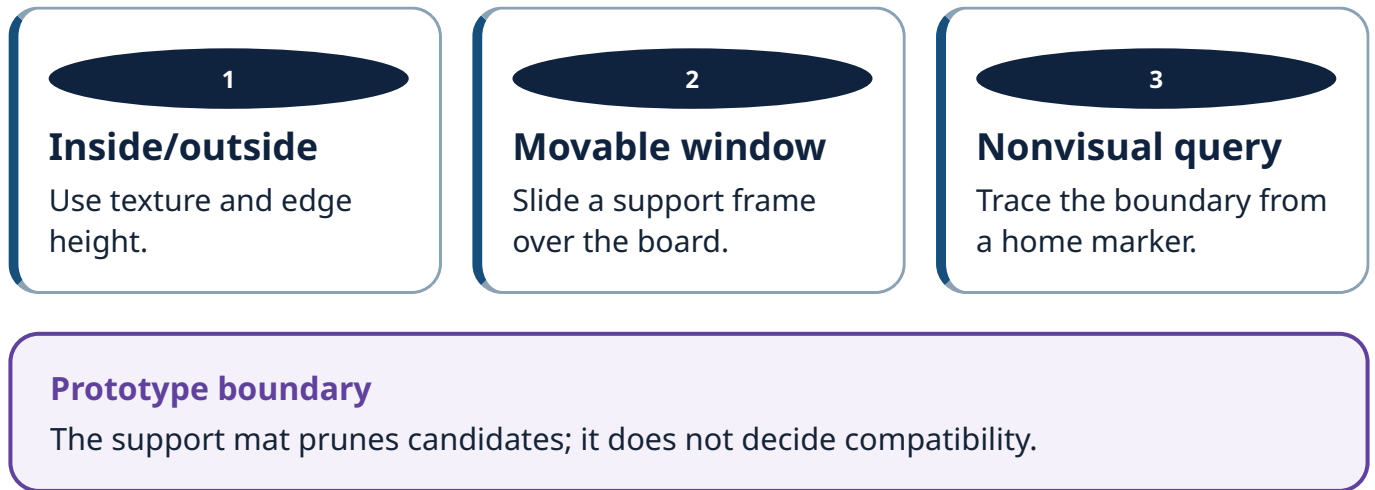
Typed sockets

Socket shape teaches domain and codomain without requiring reading.



Support mat

A raised or high-contrast region shows where a rule is relevant.



Compatibility edges

Matching edge codes show whether supported pieces may connect.



Requirement 1
Shape code.



Requirement 2
Texture or notch code.



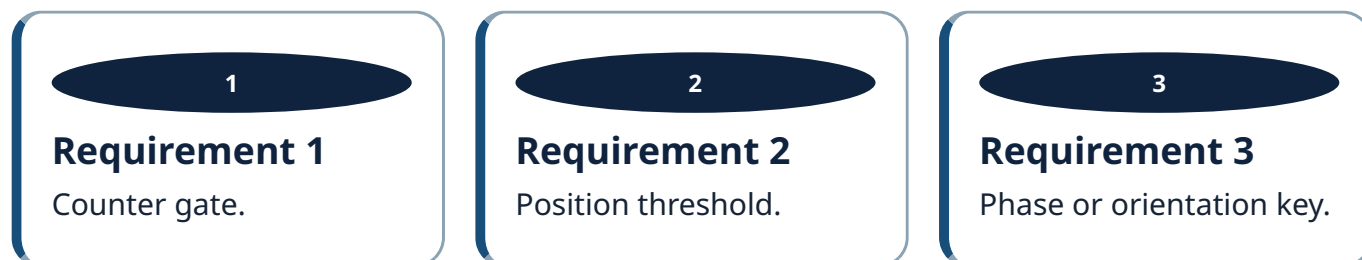
Requirement 3
Word/letter label.

Prototype boundary

A mismatch can be explored safely and explained as “not yet: these edges do not match.”

Guard gate

A gate opens only when a visible or tactile condition is reached.

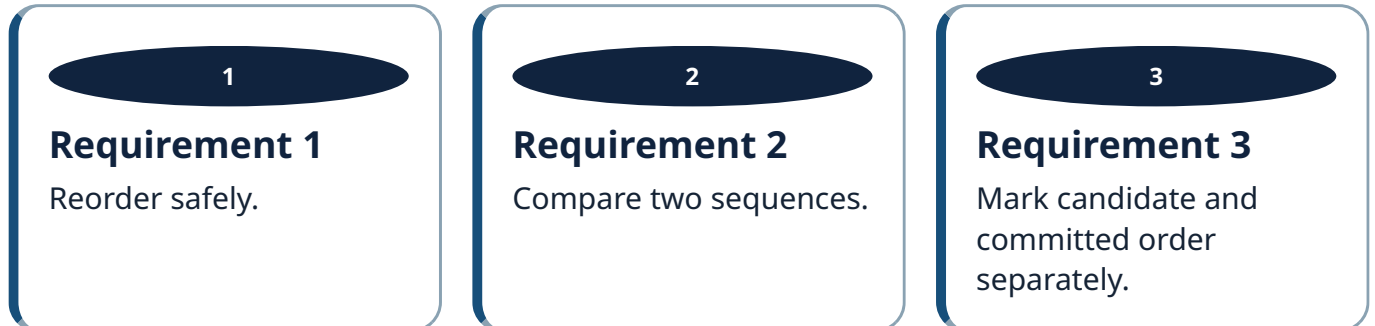


Prototype boundary

The gate has a clear candidate state and cannot snap unexpectedly onto a child's hand.

Operation-order rail

A rail makes sequence physical: cards are placed left to right and evaluated in that order.



Prototype boundary

MOVE then ROTATE and ROTATE then MOVE use the same cards but different rails.

Connector hinge

A hinge tile can join two pieces, pivot, or select between routes.

1

Requirement 1

Zero magnitude, real structure.

2

Requirement 2

Distinct pivot and connector forms.

3

Requirement 3

Guarded flip rather than loose accidental change.

Prototype boundary

The connector teaches that structural role is not the same as numerical size.

Topology track

A track can wrap, flip orientation or move between sheets while keeping the route continuous.

1**Requirement 1**

Raised arrows.

2**Requirement 2**

Orientation token.

3**Requirement 3**

Entry and exit ports with matching IDs.

Prototype boundary

The model is a routing abstraction, not a claim that a physical toy is a literal Klein bottle.

Sheet overlays

Transparent, opaque or tactile overlays allow two states to share a coordinate while remaining distinct.

1**Requirement 1**

Sheet A and B labels.

2**Requirement 2**

Compatibility windows.

3**Requirement 3**

Nontransparent option
for children who need
lower visual complexity.

Prototype boundary

A child can compare “same place” with “same identity” and “allowed to interact.”

Winding counter

A bead, dial or click wheel remembers how many loops occurred after phase repeats.

1**Requirement 1**

One tactile click per loop.

2**Requirement 2**

Parity mark for odd/
even.

3**Requirement 3**

Reset only by explicit
rule.

Prototype boundary

Winding is lineage, not hidden score.

Constraint bars

Bars, clips or rails limit legal movement directions.

1

Requirement 1

Add a constraint.

2

Requirement 2

Remove one declared shackle.

3

Requirement 3

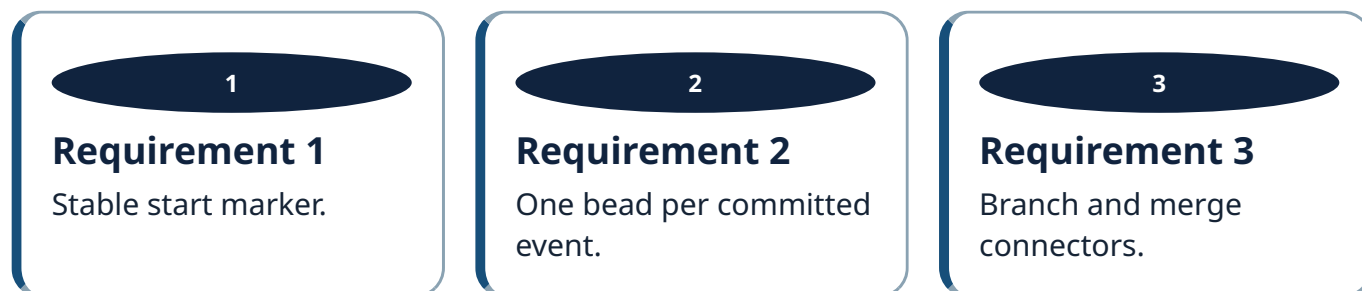
Compare movement before and after.

Prototype boundary

Use low-force parts and avoid pinch points. Constraint release does not demonstrate chaos by itself.

Replay beads and lineage chain

Each accepted change adds a bead or link with a rule code.



Prototype boundary

A child can replay by moving along the chain; rejected candidates use a separate temporary tray.

Object cues and tactile schedule

Objects announce the activity and sequence before play.

1

Requirement 1

Rule tile = rule game.

2

Requirement 2

Soft ring = break.

3

Requirement 3

Replay bead = review.

Prototype boundary

Meanings are agreed with the child and remain consistent across settings.

Low-force and mounting options

Pieces work on a table, lap tray, vertical board, non-slip mat or mounted switch surface.

1

Requirement 1

Large grasp area.

2

Requirement 2

Low insertion force.

3

Requirement 3

Magnetic and magnet-free prototypes.

Prototype boundary

No child must stabilize the whole board while operating a piece.

Tactile and large-print variants

Offer large pieces, braille-ready labels, tactile line diagrams, strong contrast and simplified layouts.

1

Requirement 1

Tactile symbols require user testing.

2

Requirement 2

Braille placement does not cover other cues.

3

Requirement 3

Provide orientation marks.

Prototype boundary

Braille and tactile graphics require qualified review; placeholders are not validated codes.

Quiet and magnet-free variants

A quiet kit uses soft stops and nonclicking markers. A magnet-free kit uses keyed mechanical connections.

1

Requirement 1

No surprise snap.

2

Requirement 2

No loose electronic sound.

3

Requirement 3

Material choices documented.

Prototype boundary

Variants preserve the same connection logic and reward.

Household prototype route

Early learning prototypes can use card, foam, hook-and-loop, string, bottle caps and textured tape.

1

Requirement 1

Use oversized parts.

2

Requirement 2

Round sharp corners.

3

Requirement 3

Supervise and inspect wear.

Prototype boundary

Household prototypes are adult-prepared learning aids, not certified toys.

Product development placeholders

Every toy concept includes spaces for age range, part size, material, force, cleaning, tactile code, mounting, packaging and test evidence.



Physical safety and regulatory boundary

The substrate can define a learning mechanic, but it cannot certify a product.

1

Requirement 1

Choking and ingestion.

2

Requirement 2

Pinch, entrapment and sharp edges.

3

Requirement 3

Magnets, batteries and cords.

4

Requirement 4

Flammability and chemical safety.

5

Requirement 5

Cleaning and infection control.

6

Requirement 6

Age grading and supervision.

Prototype boundary

Qualified toy-safety, accessibility, clinical and regulatory review is required before sale or deployment.

A ten-minute lesson

Goal: experience support and compatibility without reading or speed.

1

1 minute

Show STOP and the child's answer route.

2

2 minutes

Place two rule tiles and a support mat.

3

3 minutes

Find which tile is inside support.

4

2 minutes

Test whether edges match.

5

2 minutes

Preview, confirm and replay one change.

Facilitator rule

Finish while the child still has agency and energy; a complete curriculum is not required in one session.

A thirty-minute lesson

Goal: build a complete six-question trace with breaks.

1

Welcome

Access check and sensory forecast.

2

Explore

Definitions, dependencies and support.

3

Build

Compatibility, guard and candidate.

4

Commit

Confirm and change.

5

Replay

Lineage chain and reflection.

Facilitator rule

Offer micro-breaks and an early FINISHED option at every transition.

Cooperative classroom roles

Roles let children contribute through different strengths without ranking them.

1

Observer

Notices objects and support.

2

Matcher

Checks compatibility.

3

Guard Keeper

Watches the boundary.

4

Committer

Confirms the group choice.

5

Replay Guide

Retells or reconstructs changes.

Facilitator rule

Roles rotate only with consent; every role can be performed by speech, AAC, touch, switch, gaze or partner-assisted choice.

Family play

Use ordinary routines to reveal substrate ideas.

1

WHERE

Which shelf can hold this object?

2

MATCH

Which lid fits?

3

WHEN

When is the cup full?

4

CHANGE

What happens after the timer?

5

REMEMBER

Put one bead for each accepted change.

Facilitator rule

Keep the activity playful and stop before it becomes a lesson demand.

Accessible assessment

Assess the concept through multiple expressions, not reading speed, speech or fine-motor performance.

1

Show

Arrange tiles or point.

2

Tell

Speak, sign or use AAC.

3

Do

Choose a candidate and repair it.

4

Replay

Explain what changed in any channel.

Facilitator rule

Record evidence of understanding separately from access effort and adult support.

Child co-design

Children shape rules, access routes, language, symbols, sensory levels and repair behavior.

1

Accessible invitation.

Real choices and the right to decline.

2

Prototype in the child's route.

Pay or recognize contribution fairly.

3

Requirement 3

Show what changed because of feedback.

Facilitator rule

Co-design is not a one-time usability test after decisions are fixed.

Facilitator behavior

The adult makes access possible without becoming the player.

1

Requirement 1

Offer, do not force.

2

Requirement 2

Wait, do not fill silence.

3

Requirement 3

Confirm, do not assume.

4

Requirement 4

Describe, do not quiz.

5

Requirement 5

Repair, do not blame.

6

Requirement 6

Stop when the child stops.

Facilitator rule

Training includes communication partner fidelity and touch-consent practice.

Safeguarding and assent

Participation is voluntary and continuously revocable.

1

Explain activity and data.

Provide STOP and exit before consent.

2

Watch for child-specific distress or withdrawal signals.

Never withhold ordinary support to obtain participation.

Facilitator rule

Research or evaluation requires appropriate ethical, legal and organizational review.

Combined-access test matrix

Test realistic combinations instead of isolated checkboxes.

1

Requirement 1

Blind + switch.

2

Requirement 2

Low vision + gaze.

3

Requirement 3

AAC + partner scan.

4

Requirement 4

Deafblind + fatigue.

5

Requirement 5

Reduced motion + screen reader.

6

Requirement 6

One hand + captions + cognitive simplification.

Facilitator rule

Record route discovery time, successful stop, confirmation, repair, return and concept parity.

Child-defined metrics

Ask whether the experience worked for the child.

1

Requirement 1

Could I find a route?

2

Requirement 2

Could I stop?

3

Requirement 3

Did the game understand me?

4

Requirement 4

Could I fix a misunderstanding?

5

Requirement 5

Could I learn the same idea?

6

Requirement 6

Could I return comfortably?

Facilitator rule

Metrics are not used to rank children; they evaluate the design.

Release evidence pack

Every release carries traceable accessibility evidence.

1

Requirement 1

Requirement map.

2

Requirement 2

Automated lint report.

3

Requirement 3

Assistive-technology test cases.

4

Requirement 4

Child and family feedback.

5

Requirement 5

Known limitations and mitigations.

6

Requirement 6

Defect owner and retest date.

Facilitator rule

Standards crosswalks support evidence but do not replace participatory testing.

Roadmap and kill criteria

Stop or simplify a feature when it cannot preserve child agency, concept parity or safe use.

1

Kill drag-only, speech-only and color-only gates.

Kill unconfirmed partner control.

2

Kill inaccessible STOP, help, settings or return.

Kill unequal reward or segregated endings.

3

Requirement 3

Kill privacy-invasive gaze or emotion tracking by default.

Facilitator rule

A smaller accessible mechanic is better than a larger inaccessible claim.

Printable core communication board

Point, look, touch, sign, scan or have a partner name each cell. Keep this board visible throughout play.

STOP**GO****WAIT****HELP****AGAIN****FINISHED****YES****NO****NOT SURE****NOT THAT****CHANGED MIND****BREAK****QUIET****DIM****STILL****BIGGER****READ****TOUCH****SWITCH ROUTE**

Prototype sheet - print at 100% or enlarge. Add tactile or symbol supports only through user-tested conventions.

Printable YES / NO / NOT SURE / CHANGED MIND

Four equal-size choices. Do not place YES in a visually privileged position.

YES

NO

NOT SURE

CHANGED MIND

Printable STOP / HELP / WAIT / BREAK

Safety and control words remain available before, during and after every action.

STOP

HELP

WAIT

BREAK

Prototype sheet - print at 100% or enlarge. Add tactile or symbol supports only through user-tested conventions.

Printable **FIRST / THEN / NEXT / FINISHED**

Place one object, word or symbol under each heading. The child may reorder or remove steps.

FIRST

THEN

NEXT

FINISHED

Printable sensory forecast

Mark NONE, LOW, MEDIUM or HIGH and whether it can be changed, skipped or replaced.

SOUND

LIGHT

MOTION

TOUCH

READING

TIME

PEOPLE

BREAKS

Printable access-route chooser

Choose one or more routes for this session. A route can be changed without losing the puzzle.

LOOK**LISTEN****TOUCH****READ****SPEAK****SIGN****AAC****SWITCH****GAZE****HEAD****KEYS****PARTNER**

Printable My Access Passport

Share only what the child chooses. Do not add diagnosis labels unless the child or family explicitly wants them included.

MY YES

MY NO

MY STOP

MY NOT SURE

WHAT HELPS

WHAT TO AVOID

HOW TO WAIT

HOW TO RETURN

Printable partner scan record

Use to train and review partner fidelity. The child is not being scored.

CHOICES

SCAN ORDER

CHILD SIGNAL

CONFIRMED CHOICE

REPAIR USED

WAIT TIME

Printable six-question cards

Cut into large cards. Add tactile marks or object cues only after user testing.

**WHERE?
SUPPORT**

**MATCH?
COMPATIBILITY**

**WHEN?
GUARD**

**CHECKED?
VERIFIED**

**CHANGE
TRANSITION**

**REMEMBER
LINEAGE**

Prototype sheet - print at 100% or enlarge. Add tactile or symbol supports only through user-tested conventions.

Printable blank rule tile

Use large print and one idea per tile. Add a distinct border pattern and typed socket sketch.

RULE NAME

LETTER CODE

WHAT IT NEEDS

WHEN IT RUNS

WHAT IT CHANGES

HOW IT IS REMEMBERED

Printable dependency cable labels

Attach labels to string, pipe cleaners or card connectors. Keep loose-part safety in mind.

NEEDS

PROVIDES

BEFORE

AFTER

SAME TYPE

DOES NOT MATCH

OPTIONAL

REQUIRED

Prototype sheet - print at 100% or enlarge. Add tactile or symbol supports only through user-tested conventions.

Printable operation-order rail

Place rule cards in order. Compare two rails without changing the card identities.

1 FIRST

2 SECOND

3 THIRD

4 FOURTH

5 PREVIEW

6 COMMIT

Prototype sheet - print at 100% or enlarge. Add tactile or symbol supports only through user-tested conventions.

Printable candidate and real-world mat

Candidate pieces never move to REAL WORLD without explicit confirmation.

TRY / CANDIDATE

CHECKS

CONFIRM

REAL WORLD

CANCEL

REPLAY

Prototype sheet - print at 100% or enlarge. Add tactile or symbol supports only through user-tested conventions.

Printable repair ladder

Use after a misunderstanding. Never require the child to apologize for access failure.

STOP

**WHAT DID YOU THINK I
CHOSE?**

YES / NO / NOT THAT

SHOW AGAIN

CHOOSE ANOTHER ROUTE

CONFIRM

Printable topology track

Draw or build a track. Use arrows and tactile orientation marks. Record the wrap as an event.

ENTRY

WRAP

FLIP

SHEET A

SHEET B

EXIT

Prototype sheet - print at 100% or enlarge. Add tactile or symbol supports only through user-tested conventions.

Printable winding and replay beads

Thread or arrange large safe pieces. Keep winding count separate from phase position.

START

LOOP 1

LOOP 2

LOOP 3

BRANCH

MERGE

CHECKPOINT

FINISH

Prototype sheet - print at 100% or enlarge. Add tactile or symbol supports only through user-tested conventions.

Printable cooperative role cards

Any role can use any communication or input route. A child may pass or share a role.

OBSERVER

MATCHER

GUARD KEEPER

COMMITTER

REPLAY GUIDE

ACCESS CHECKER

Printable return-to-play card

Use after a break, device loss or interrupted session. Do not test memory.

YOU WERE HERE

LAST ACCEPTED CHANGE

CURRENT CANDIDATE

YOUR ACCESS ROUTE

CONTINUE

REVIEW

LEAVE

Source map and guidance crosswalk



This edition combines the suitable substrate versions without changing their evidence boundaries.

UGTS-GN 1.1

Canonical support, compatibility, guard, event, transition and lineage architecture; precision, memory and claim boundaries.

UGTS-KC 2.0

Patterns, fields, topology descriptors, kinematics, bounded dynamics and hybrid-event classification.

KC Two Hands 3.0

Stable scene identity, typed interaction, proposals, deterministic commit, checkpoints and replay.

UGTS-KC 3.6

First-class definitions, content addresses, dependency graphs, hinges and explicit operation order.

SCLP 3.6.2

Guard intervals, winding, reflective topology, constraint release, bounded grammar, finite keys and compression audit.

KC Elizabeth 3.9

Vector assets, named multi-device input, deterministic fixed-step game world, persistence and offline HTML5 delivery.

SARA 3.6.3

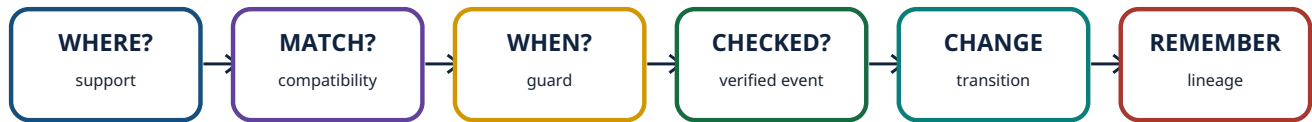
Optional older-learner public/private/authorization lesson. Secret search and transaction capabilities remain excluded.

Accessibility guidance crosswalk

Design and testing are informed by W3C WCAG 2.2 and WCAG2ICT, CAST Universal Design for Learning Guidelines 3.0, Xbox Accessibility Guidelines, UNICEF accessible digital learning work, and National Center on Deafblindness principles. These are guidance sources, not a claim of certification.

w3.org/TR/WCAG22/ | w3.org/TR/wcag2ict-22/ | udlguidelines.cast.org | learn.microsoft.com/gaming/accessibility/xbox-accessibility-guidelines | unicef.org | nationaldb.org

Formal decision and validation boundary



Definition Breaker DB-2.0 is ready as a child-directed accessibility, learning and prototype standard.

What this document establishes

- A complete eight-stage access envelope.
- Child-controlled communication, input, sensory and return routes.
- A child-to-expert learning bridge for the suitable UGTS substrate versions.
- Digital implementation contracts, physical toy scaffolds, teaching protocols and printable prototypes.

What still requires real-world validation

- Participatory testing with diverse children and families.
- Independent PDF/UA and assistive-technology review.
- Educational, clinical and safeguarding review where applicable.
- Braille, tactile-symbol and sign-language validation.
- Toy safety, age grading, manufacturing and regulatory assessment.

FORMAL DECISION

Accessibility changes the route, not the child's authority or the substrate concept. Every essential action must be perceivable, understandable, operable, confirmable, repairable, pausable and resumable through at least one route that works for the child in that session. The same verified proposal, transition and lineage path remains authoritative.

WHERE? MATCH? WHEN? CHECKED? CHANGE. REMEMBER.
The child decides how to enter the path.